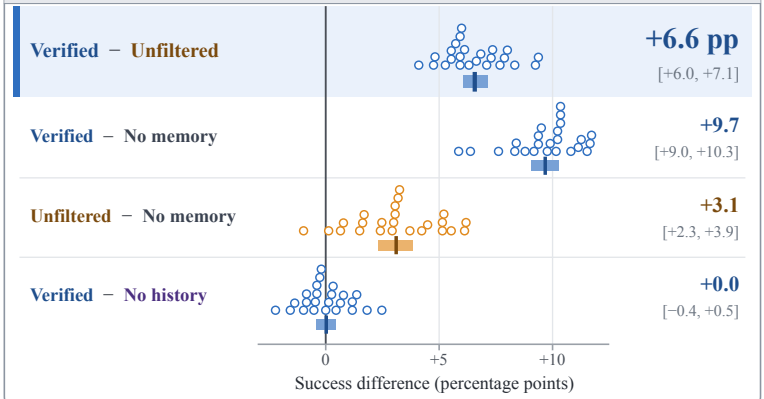
Policy ■ No memory — Unfiltered — Verified — No history

memory budget (nothing stored) · unchecked records · checked + 25% history · checked records only

SYNTHETIC DATA — constructed demonstration, not results from real agents

(a) Paired success differences

dot: one seed block (n = 24) · tick, bar: mean, 95% CI



(b) Gain by task and budget

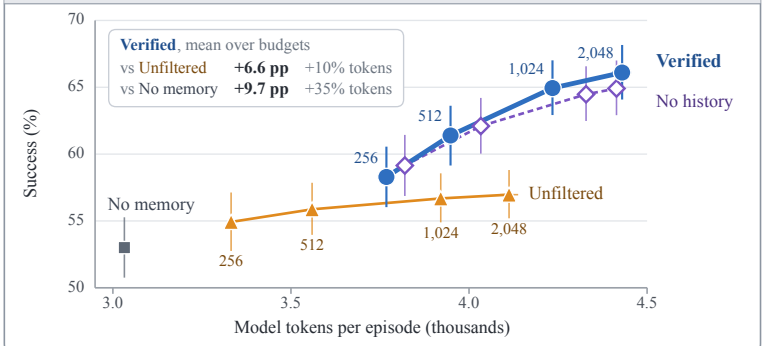
Verified − Unfiltered

	Memory budget (tokens) 256	512	1,024	2,048	mean
Fact recall	+0.3	+0.5	+1.3	+1.9	+1.0
Code repair	+4.1	+6.9	+9.9	+11.1	+8.0
Tool routing	+5.4	+8.6	+11.6	+12.5	+9.5
Long horizon	+3.6	+6.0	+10.2	+11.1	+7.7
mean	+3.4	+5.5	+8.3	+9.1	**+6.6**

0 — +14 pp — *95% CI includes 0*

(c) Success versus token cost

mean over 24 seed blocks · bar: 95% CI



(d) Rejection-history diagnostic

